

Supplementary Tables

Ancestry sorting, genomic architecture, and climate association

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Tables are generated directly from the analysis outputs. The principal parameter settings are given in the Materials and Methods. All analyses are descriptive and conditional on the sampled populations and quality gates.

Table S1: Genome-wide sort-class counts among parent-polymorphic SNPs (pooled parental $\text{MAF} \geq 0.15$; 661,312 SNPs). Percentages are among the 655,537 loci with a defined parental orientation.

Sort class	SNPs	% of oriented
Unsorted	445,520	67.96
Unidirectional (<i>polycтена</i>)	114,712	17.50
Unidirectional (<i>aquilonia</i>)	94,394	14.40
Bidirectional	911	0.14
No defined parental orientation (excluded)	5,775	—

Table S2: Directional-predictability test (two-sided exact binomial vs $p = 0.5$; units near-fixed in ≥ 5 populations; Benjamini–Hochberg adjusted, $q < 0.05$).

Level	Tested	Significant	%
SNP	554,160	311,440	56.2
LD-reduced unit	111,184	108,548	97.6

Tests are conditional and exploratory. The much higher significant fraction at the unit level reflects the parental-polymorphism gate and the larger number of near-fixed populations required, not stronger evidence per locus.

Table S3: Aggregation of marker-level sorting by cluster size (clusters entering because they contained ≥ 1 individually sorted marker). The comparison is conditional and does not estimate the genome-wide unsorted fraction.

Cluster size (markers)	Clusters	% unsorted after aggregation
1	77,761	0.0
2-4	24,265	20.9
5-9	7,412	68.5
10-49	7,601	81.4
50-199	344	65.1
200+	63	60.3

Dilution check: among the 16,597 multi-marker clusters unsorted after aggregation, consensus fidelity remained high (median $\text{score}_{\text{MLG}} = 0.856$); only 7 fell below the 0.80 fidelity floor, so the loss of the aggregate call reflects genuine absence of a joint signal rather than averaging artefact.

Table S4: Genome-wide rank correlations at analysed SNPs. The diagnostic index is essentially orthogonal to recombination and to pooled parental MAF, and tracks reduced parental expected heterozygosity rather than elevated between-parent difference.

Variable	Variable	Spearman ρ
Diagnostic index	Recombination rate	-0.03
Diagnostic index	Pooled parental MAF	-0.02
Diagnostic index	Parental expected heterozygosity (π)	-0.46
Diagnostic index	Between-parent allelic difference (d_{xy})	+0.13

Table S5: Sorting magnitude (prop_fixed) at the LD-reduced-unit level (linear model). Cluster size is excluded as a potential collider.

Predictor	Coef.	SE	t	P
Recombination (std. log)	0.052	0.000	116.7	< 0.001
Diagnostic index (std.)	0.027	0.000	61.0	< 0.001

Table S6: Direction of sorting (logistic model of *aquilonia*-directed near-fixation among unidirectional units). Coefficients are associations, interpreted descriptively.

Predictor	Coef.	SE	z	P
Diagnostic index (std.)	1.462	0.012	125.2	< 0.001
Recombination (std. log)	-0.091	0.012	-7.6	< 0.001
Pooled parental MAF (std.)	-0.060	0.011	-5.6	< 0.001
Cluster size (std. log)	0.040	0.009	4.5	< 0.001

Table S7: Diagnostic-index enrichment: percentage-point change in ancestry-diagnostic content per +10 percentage-point increase in the cluster climate-significance rate (primary configuration: Åland-excluded, fixed- Ω ; cluster-level linear model with log cluster size).

Axis	Stratum	Effect (pp / +10pp)	P
PC1	all	+0.54	0.287
PC1	≥ 20	+1.86	0.040
PC1	≥ 50	+4.46	< 0.001
PC2	all	-0.33	0.472
PC2	≥ 20	+1.82	0.085
PC2	≥ 50	+3.72	0.013

Table S8: Sorting-climate overlap: cluster-level logistic odds ratio for directional sorting per unit significance rate (primary configuration), showing no association.

Axis	Stratum	Odds ratio	P
PC1	all	1.25	0.031
PC1	≥ 20	1.37	0.085
PC1	≥ 50	0.99	0.978
PC2	all	1.16	0.148
PC2	≥ 20	1.77	0.035
PC2	≥ 50	1.02	0.957